

Decentralized Multi-Robot Path Planning with Adaptive Diffusion Convolution Mechanism

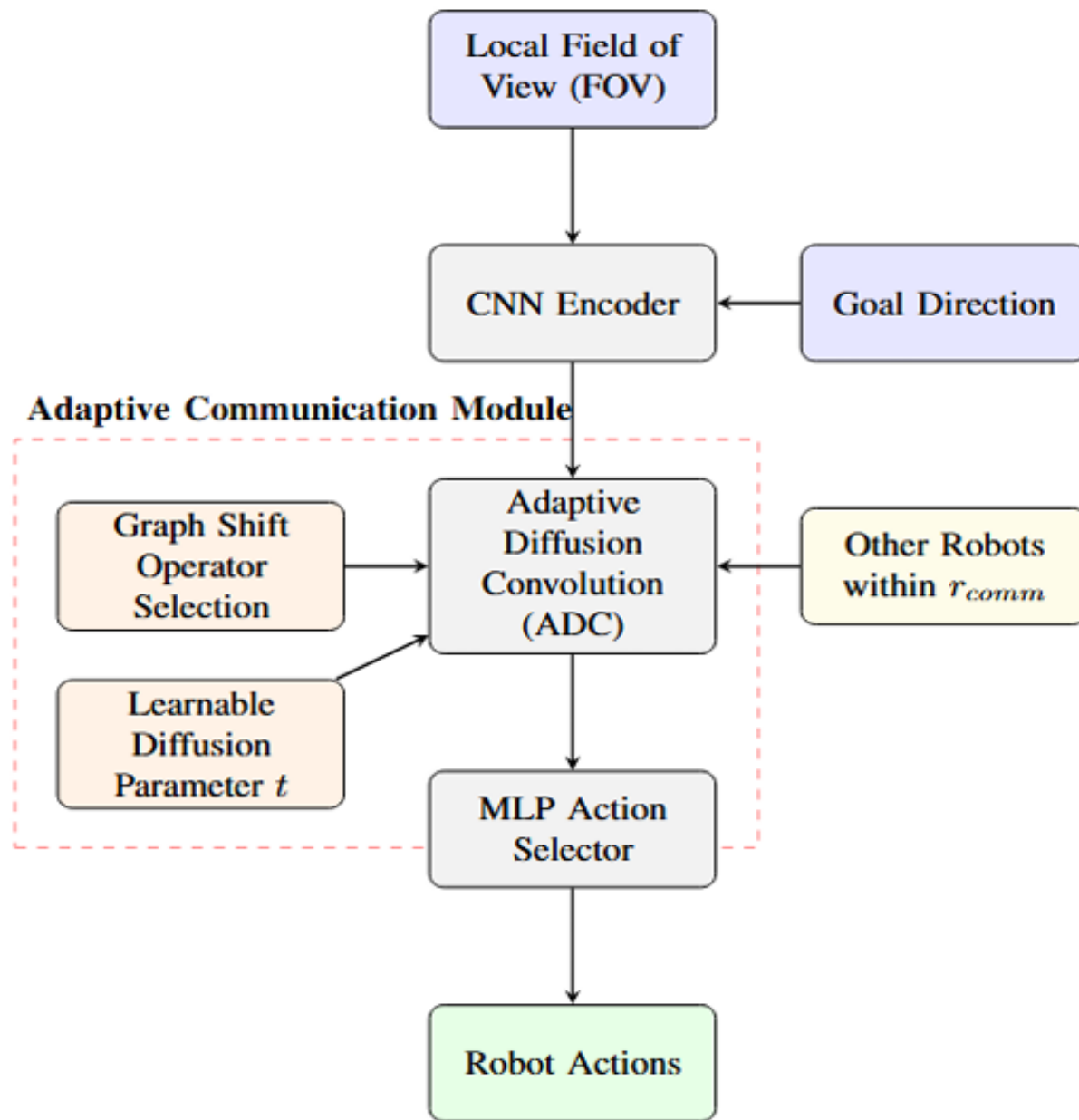


Fig. 1. Architecture of the ADC-augmented MRPP framework

Key Points-:

- Reason for using this over fixed-GNN (deadlock problem and energy problem)
- The ADC (decentralized method) mechanism uses a heat filter $\theta_k(t) = e^{-tL}$, where L is the laplacian and t is the diffusion time.
- L consists of two parts Adjacency Matrix and Degree Matrix
- t is trained again and again using backpropagation through Imitation Learning (here CBS algorithm) and here in this paper it is used to effectively control the communication radius of agents
- This heat filter is working as a low-pass filter (it tries to diffuse the different parameters)
- Instead of using exponential it uses Taylor series
- Coming from CNN through ADC, the new feature vectors are sent to MLP (Multi-layer perceptron) which decides actions for agent

TABLE I

PERFORMANCE OF MODELS TRAINED ON 10% OBSTACLE DENSITY

Test Density	Model	SR (%)	AM	SOC	Params
10%	GCN (K=1)	74.53	10.30	65.54	39.7k
	GCN (K=2)	80.95	10.46	57.47	47.9k
	GCN (K=3)	79.09	10.41	59.98	56.1k
	GCN (K=4)	80.43	10.51	57.88	64.3k
	ADC-Main	75.47	10.34	64.06	39.7k
	ADC-FixedT	81.16	10.55	57.64	39.7k
	ADC-K1	76.50	10.38	63.14	39.7k
20%	GCN (K=1)	50.87	10.39	102.96	39.7k
	GCN (K=2)	57.04	10.73	95.04	47.9k
	GCN (K=3)	58.21	10.77	93.78	56.1k
	GCN (K=4)	56.34	10.69	95.76	64.3k
	ADC-Main	55.88	10.69	96.07	39.7k
	ADC-FixedT	55.53	10.95	97.88	39.7k
	ADC-K1	53.90	10.50	99.19	39.7k
30%	GCN (K=1)	38.38	10.26	132.13	39.7k
	GCN (K=2)	36.93	10.35	134.12	47.9k
	GCN (K=3)	37.34	10.18	133.99	56.1k
	GCN (K=4)	39.42	10.46	131.33	64.3k
	ADC-Main	39.42	10.59	130.01	39.7k
	ADC-FixedT	39.21	10.87	131.99	39.7k
	ADC-K1	35.68	10.17	136.26	39.7k

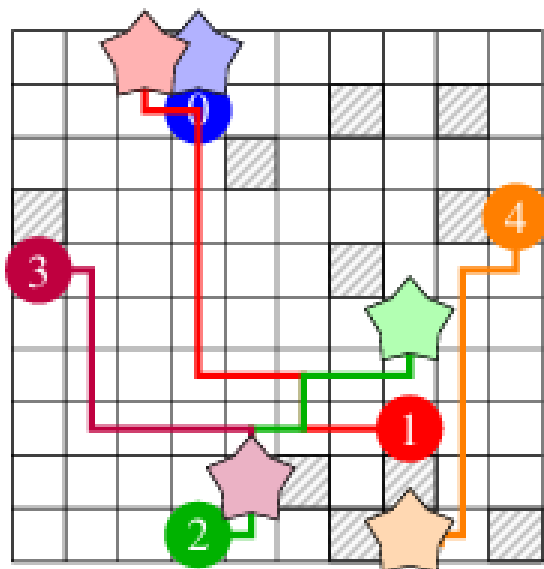
TABLE II

PERFORMANCE OF MODELS TRAINED ON 20% OBSTACLE DENSITY

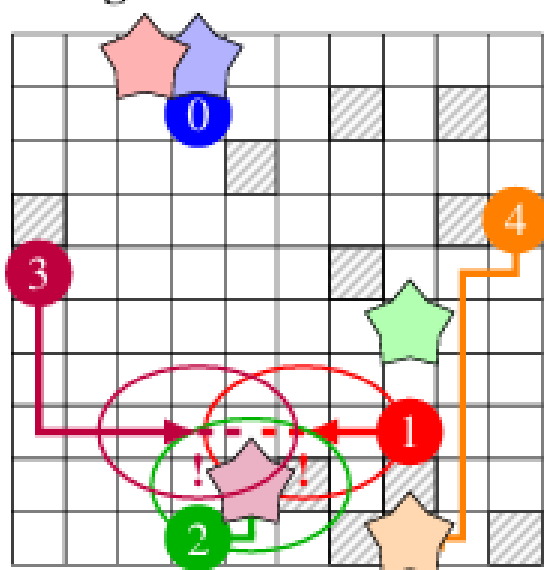
Test Density	Model	SR (%)	AM	SOC	Params
10%	GCN (K=1)	74.22	10.31	65.94	39.7k
	GCN (K=2)	78.99	10.43	60.64	47.9k
	GCN (K=3)	77.74	10.50	60.97	56.1k
	GCN (K=4)	80.85	10.46	57.72	64.3k
	ADC-Main	77.12	10.25	63.00	39.7k
	ADC-FixedT	83.02	10.57	55.77	39.7k
	ADC-K1	75.88	10.29	64.09	39.7k
20%	GCN (K=1)	55.53	10.61	96.71	39.7k
	GCN (K=2)	62.17	10.94	87.41	47.9k
	GCN (K=3)	61.00	10.94	89.32	56.1k
	GCN (K=4)	61.93	10.87	88.20	64.3k
	ADC-Main	58.56	10.70	92.43	39.7k
	ADC-FixedT	64.61	11.31	85.37	39.7k
	ADC-K1	59.25	10.65	91.30	39.7k
30%	GCN (K=1)	44.40	10.70	123.32	39.7k
	GCN (K=2)	46.68	11.03	121.52	47.9k
	GCN (K=3)	44.19	10.64	124.02	56.1k
	GCN (K=4)	46.89	10.71	120.66	64.3k
	ADC-Main	43.98	10.80	123.69	39.7k
	ADC-FixedT	46.27	11.13	121.43	39.7k
	ADC-K1	45.64	10.74	121.33	39.7k

TABLE III
PERFORMANCE OF MODELS TRAINED ON 30% OBSTACLE DENSITY

Test Density	Model	SR (%)	AM	SOC	Params
10%	GCN (K=1)	76.40	10.40	62.84	39.7k
	GCN (K=2)	77.02	10.44	62.43	47.9k
	GCN (K=3)	77.12	10.49	62.94	56.1k
	GCN (K=4)	77.43	10.55	61.53	64.3k
	ADC-Main	75.88	10.39	63.52	39.7k
	ADC-FixedT	81.47	10.69	56.70	39.7k
	ADC-K1	76.29	10.42	63.19	39.7k
20%	GCN (K=1)	58.44	10.66	93.39	39.7k
	GCN (K=2)	61.35	10.81	89.62	47.9k
	GCN (K=3)	60.42	11.00	90.74	56.1k
	GCN (K=4)	59.25	10.76	91.75	64.3k
	ADC-Main	59.02	10.72	92.14	39.7k
	ADC-FixedT	65.31	11.25	84.89	39.7k
	ADC-K1	55.41	10.62	97.03	39.7k
30%	GCN (K=1)	43.36	10.73	125.00	39.7k
	GCN (K=2)	44.19	10.85	123.30	47.9k
	GCN (K=3)	46.06	11.24	120.55	56.1k
	GCN (K=4)	47.10	10.59	119.83	64.3k
	ADC-Main	45.85	10.71	121.38	39.7k
	ADC-FixedT	47.51	11.39	119.55	39.7k
	ADC-K1	45.02	10.74	123.00	39.7k



(a) ADC: All robots successfully reach goals with coordination



(b) Fixed-K GNN ($K=3$): Circular deadlock from simultaneous moves

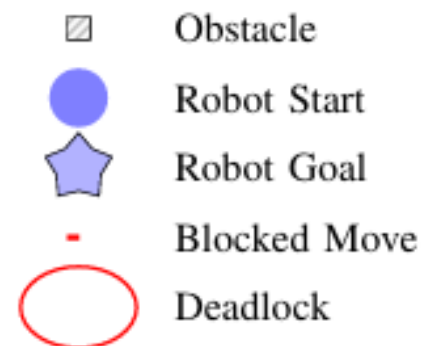


Fig. 2. Multi-robot path finding comparison. (a) ADC coordinates all robots successfully, (b) fixed- K GNN ($K=3$) fails when robots 1 and 2 simultaneously target position (4, 2), creating circular deadlock: Robot 1 stuck at (5, 2), Robot 2 at (4, 1), and Robot 3 at (3, 2)